Data science School

Deep Learning models for predicting memory impairments stages using structural and functional MRI

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Early Prediction of Alzheimer's Disease

- Alzheimer's Disease (AD) is a progressive brain disorder affecting memory and cognition.

- Early detection is crucial for improving quality of life and slowing disease progression.

- AD begins with mild symptoms and worsens over time.

Currently, there's no cure for AD ,treatments focus on managing symptoms and slowing progression.

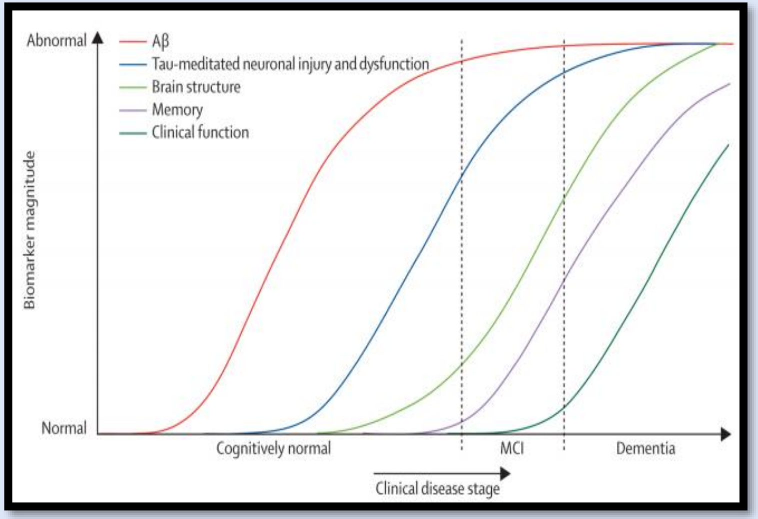
Motivation

- The motivation behind this study is to overcome the challenges associated with Alzheimer's Disease (AD) using MRI data.
- With the help of the MRI & fMRI data, we will explore normal cognitive (CN) state ,mild cognitive impairment (MCI) or Alzheimer's Disease (AD).
- The aim is to introduce an ideal Deep Learning (DL) approach for early AD detection.

What is Mild Cognitive Impairment (MCI)?

- A transitional phase between normal aging and dementia from both cognitive and neurobiological perspectives.

- Crucial to detect early in order to begin treatment.



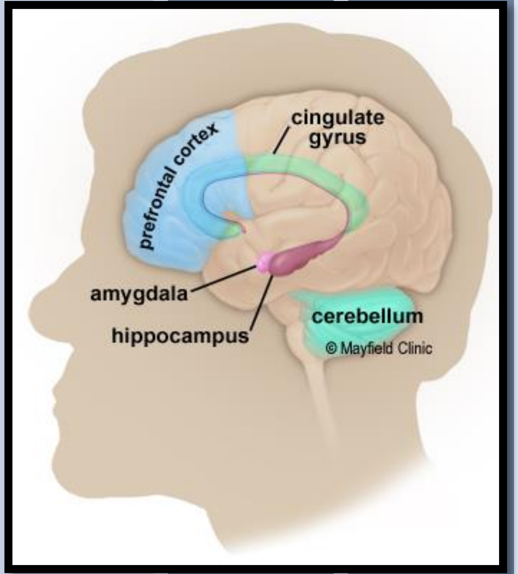
Adapted from Clifford et al. *Lancet Neurol* (2010)

Brain Anatomy – Memory

Prefrontal cortex: Responsible for higher-level thinking, such as decision-making, planning, and emotional regulation.

Hippocampus: vital for the formation and consolidation of long-term memories. It converts short-term memories into long-term memories through a process called encoding.

Cerebellum: The cerebellum is crucial for motor coordination and balance, but it also contributes to skill memory. It stores and retrieves procedural memories like motor skills and conditioned responses.



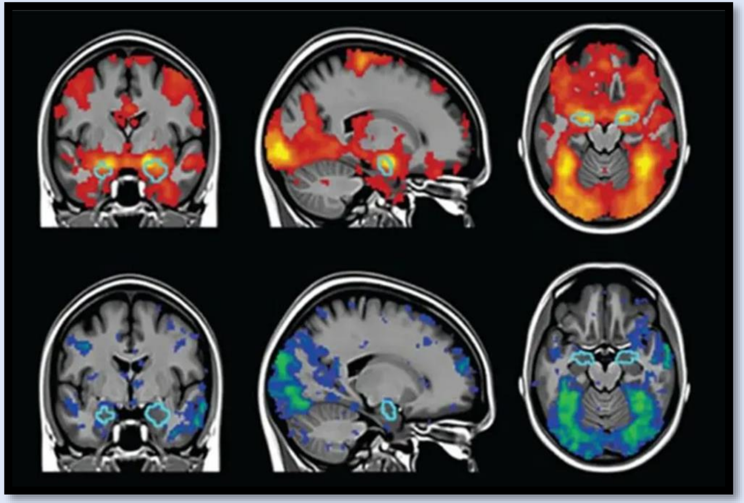
Alzheimer's disease leads to the reduction in size of the hippocampus and cerebral cortex, accompanied by an expansion of ventricles within the brain.

MRI (Magnetic Resonance Imaging)

- MRI is recognized as a vital and irreplaceable method for identifying early indicators of Alzheimer.
- The key advantages of MRI include its noninvasive nature, eliminating the need for surgical procedures and enabling comprehensive brain imaging.
- High-resolution imaging capabilities (T-weighted)

fMRI (Functional Magnetic Resonance Imaging)

fMRI Detects changes in brain activity during tasks related to memory.



Cerebral Blood Flow

Records alzheimer's affects blood flow and oxygenation patterns in the brain.

Regional Brain Activity

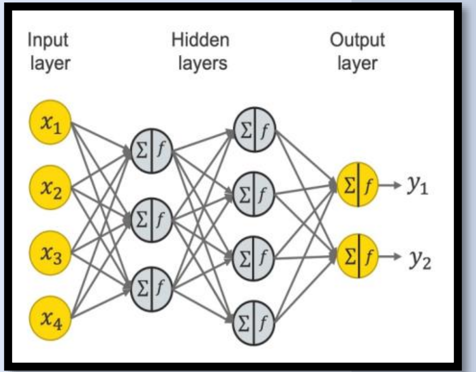
Captures detailed images of the brain during time, allowing researchers to visualize neural activity in real-time.

Functional Connectivity

fMRI can measure how different brain regions communicate and coordinate, which is disrupted in Alzheimer's disease.

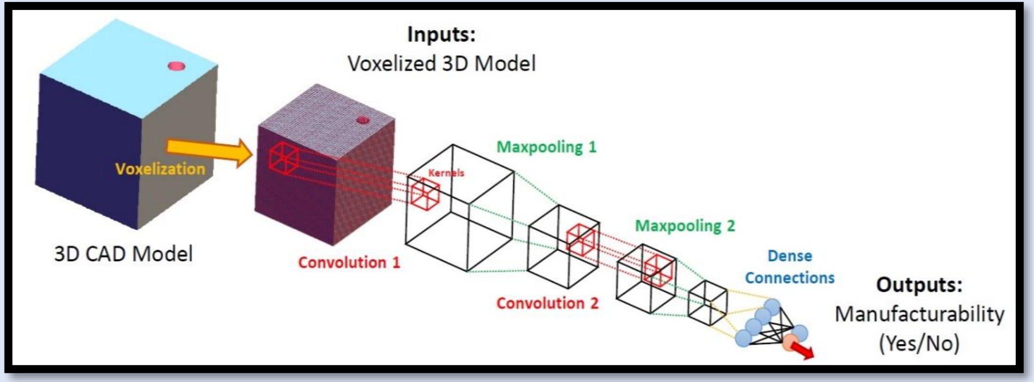
Deep Learning for Alzheimer's Detection

- **Leverages MRI and fMRI data** to understand brain changes in Alzheimer's disease.
- **Employs CNN models** to extract crucial image features for early detection and disease progression monitoring.
- **Combines MRI and fMRI** for comprehensive brain analysis.
- **Overcomes limitations of human interpretation** by identifying patterns unseen by experts.



3D Imaging for Alzheimer's Detection

- Exploits **spatial information and volumetric** analysis for feature extraction.
- Enables fusion of multiple imaging modalities (MRI, fMRI, DTI).
- Captures complex brain patterns.
- Improves Alzheimer's diagnosis and prognosis.

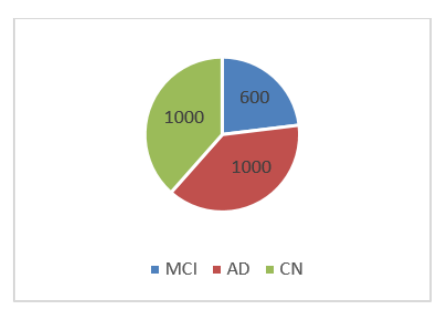


Data Set

- **Alzheimer’s Disease Neuroimaging Initiative (ADNI):** ADNI collects a vast amount of data from participants, including clinical, imaging, genetic, and biomarker data for Alzheimer's research. We **focused on 3D imaging datasets**.

- **fMRI (RS-fMRI Data):** Pre-processed by Dr. Gergo Bolla from Semmelweis University. Includes HC and MCI participants, with 4D nii files for brain activity analysis.

MRI Model



Process of Work:

Utilized **ADNI2** 3D MRI data, classified into 3 groups (CN, MCI, AD) in Nifti format.

Data files were divided into lists based on its labels.

Data Preprocessing:

Resizing: Images were resized to (90, 108, 108) due to varying shapes and memory limitations.

Data Split:

❖ Training set – 70%.
❖ Validation set – 15%.
❖ Test set – 15%.

3D Convolutional Neural Network (CNN) Model

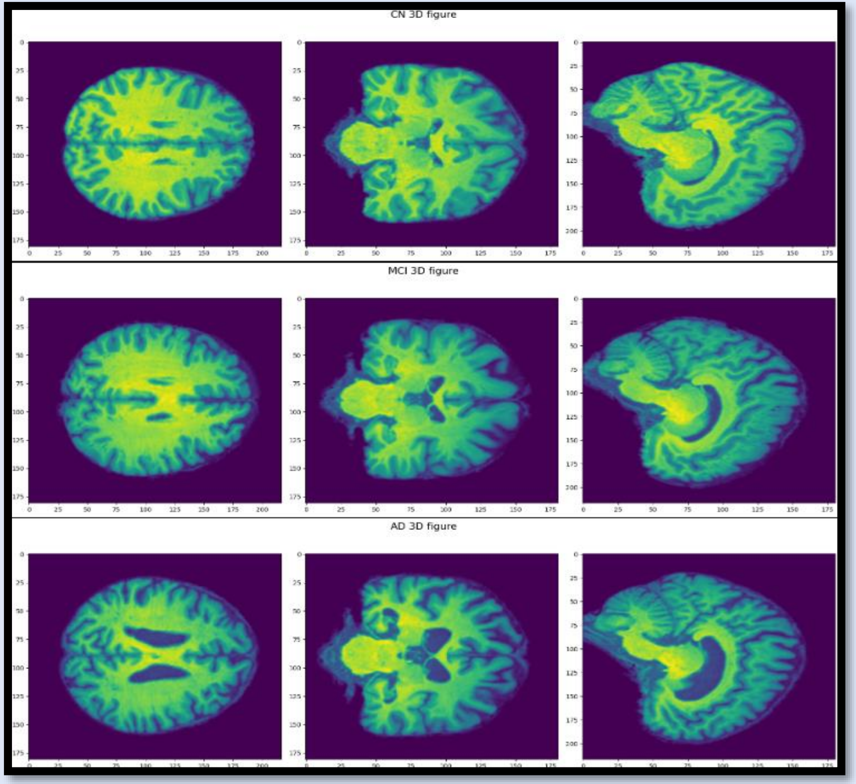
Hyperparameters:

Convolutional filters	64,32
Kernel size	3*3*3
Batch size	16
dropout	0.3
Pool size	2*2*2
epochs	200 (stopped at 175)
Learning rate	0.001
loss	"CE"
optimizer	"Adam"
patience	30

```
model = models.Sequential()

# Convolutional layers
model.add(Conv3D(64, kernel_size=(3, 3, 3), activation='relu', input_shape=(x_size, y_size, z_size,1)))
model.add(MaxPooling3D(pool_size=(2, 2, 2)))
model.add(Dropout(0.3))
model.add(Conv3D(32, kernel_size=(2, 2, 2), activation='relu'))
model.add(MaxPooling3D(pool_size=(2, 2, 2)))

model.add(Flatten())
model.add(Dropout(0.3))

# Dense layers
model.add(Dense(128, activation='relu'))
model.add(Dense(3, activation='softmax'))
```

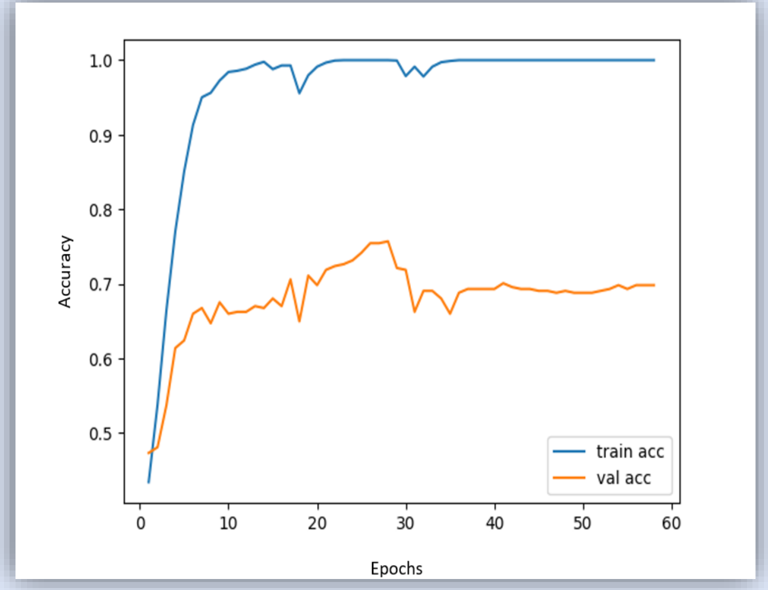
3D 3T MRI scans

MRI Scans for each label (CN, MCI, AD):

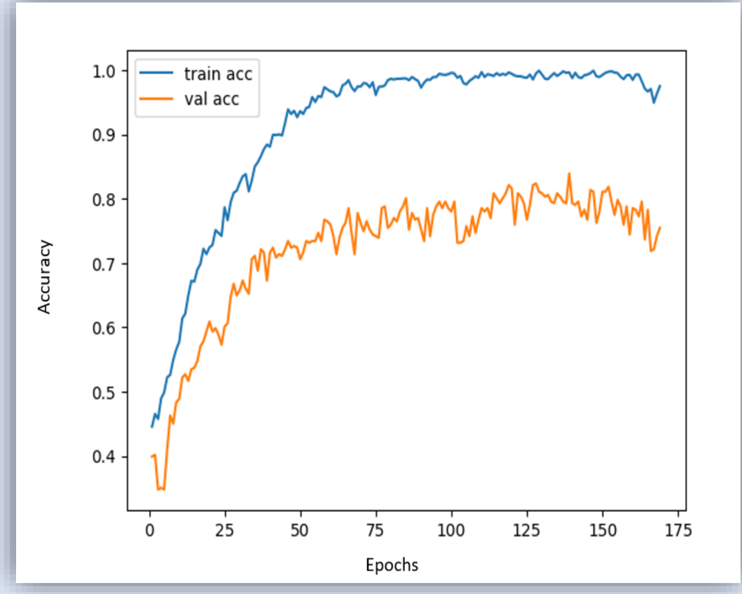


3D CNN Results

Without dropout



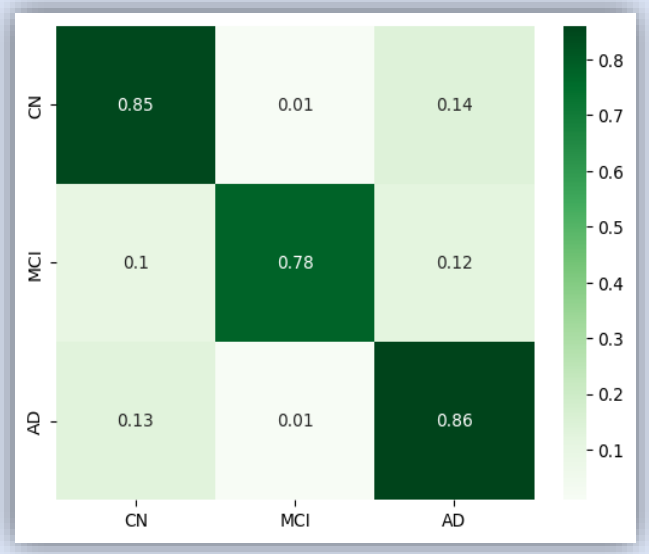
With dropout



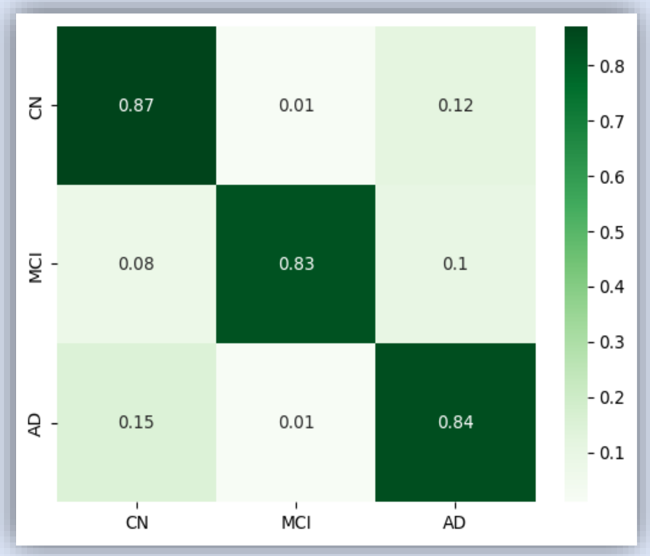
	Validation Accuracy	Test Accuracy
Without dropout model	75.7%	74.2%
With dropout model	83.88%	84.7%

3D CNN Results

With dropout on **validation set**

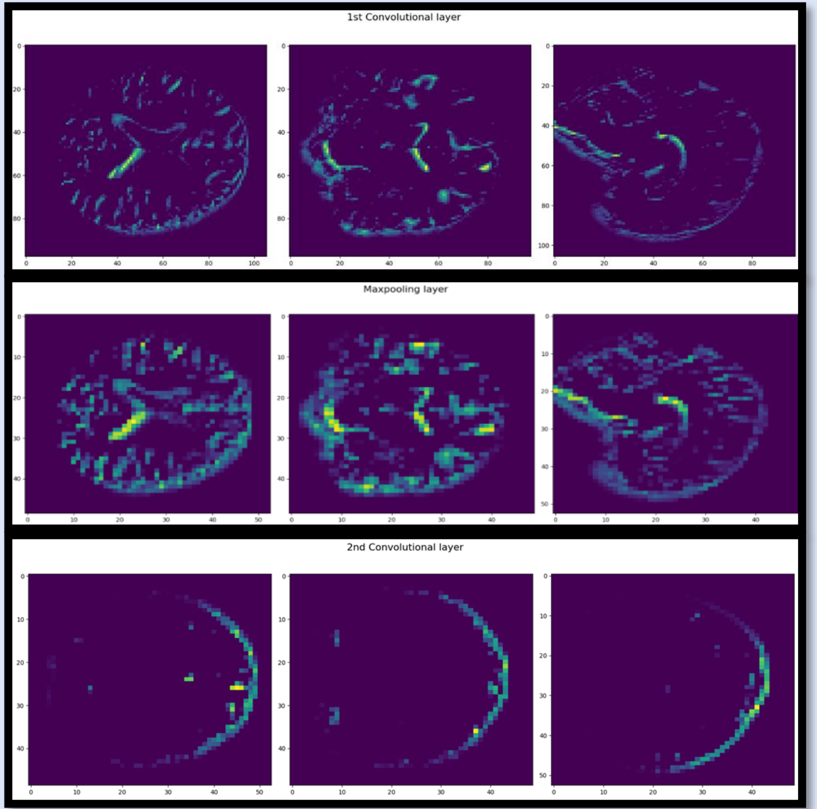


With dropout on **test set**



Model's Activations

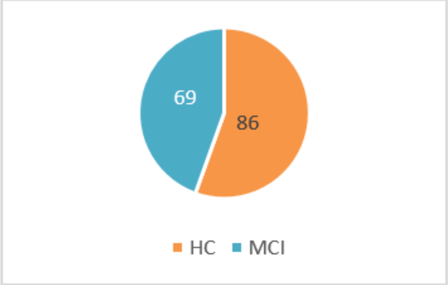
We extracted the activation patterns of the trained 3d-CNN layers, as a tool that could pinpoint to the regions involved with the progress of Alzheimer's Disease



Conclusions

- The model demonstrated relatively **high accuracy** for each class, indicating its ability to correctly classify samples belonging to CN, MCI, and AD categories.
- Generally, the model didn't tend to overfit to any one of the classes.
- The dropout model needed much **longer training**, but yielded much **better results**.
- The model manages to focus on the critical areas for classification at any image for each one of the classes.

fMRI Model



Process of Work:

Utilized preprocessed fMRI data from Dr. Bolla 86 HC, 69 MCI participants.

Data Alignment: Standardized time dimension to 140 frames across all files.

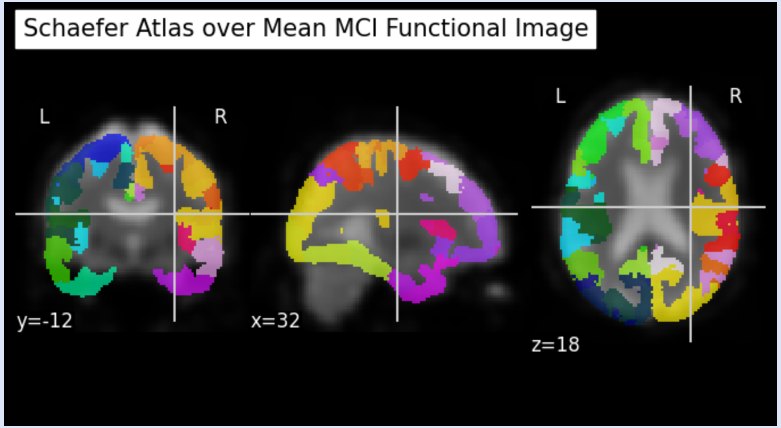
Resizing: Images were resized with Nilearn Schaefer Atlas from 3D imageing to ROIs (Region of interest) (140,91,109,91) -> (140,200).

Data Split:

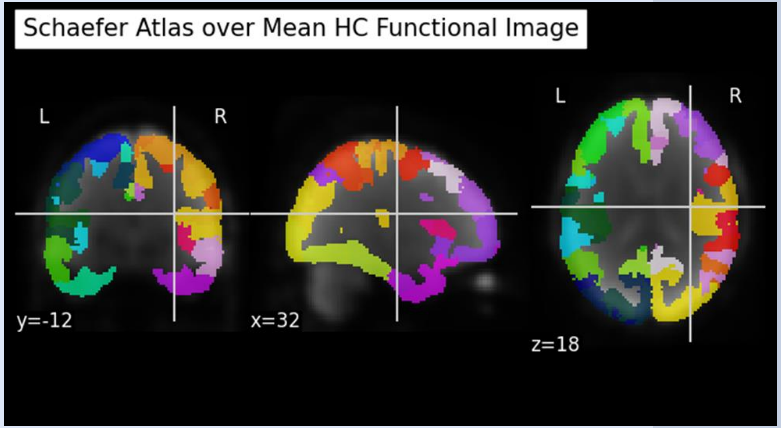
- Training set – 70%.
- Validation set – 15%.
- Test set – 15%.

fMRI Pre-processing

Schaefer Atlas on MCI



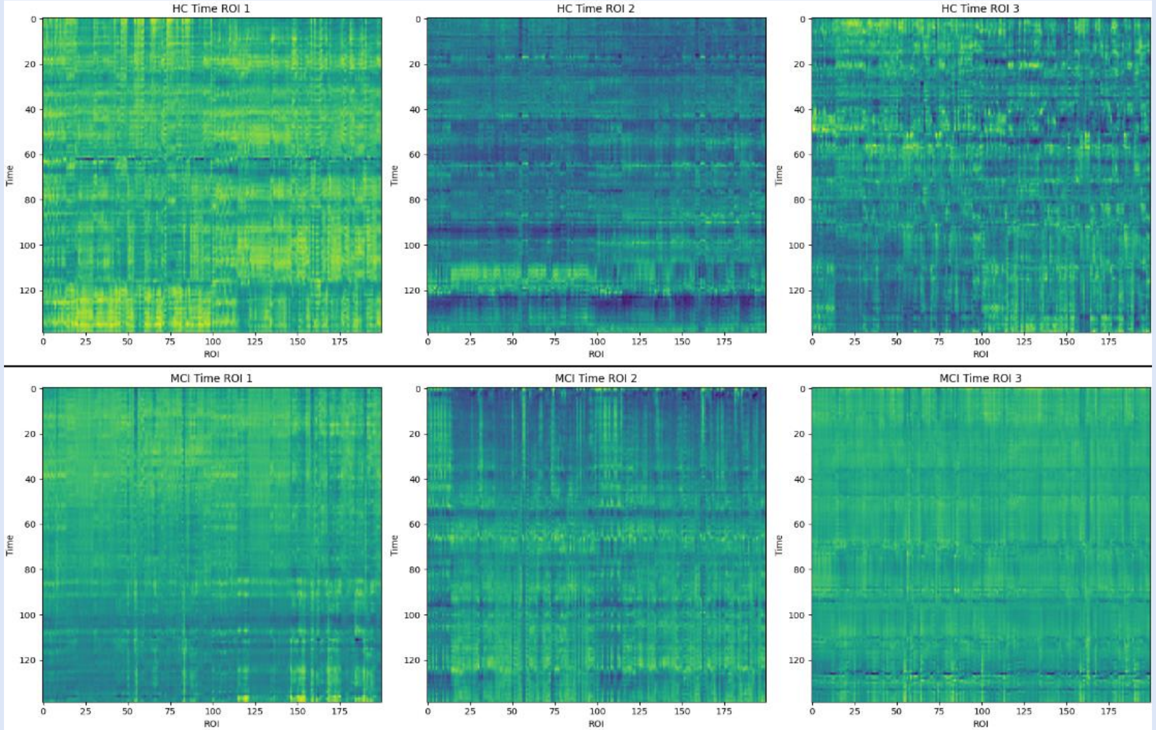
Schaefer Atlas on HC



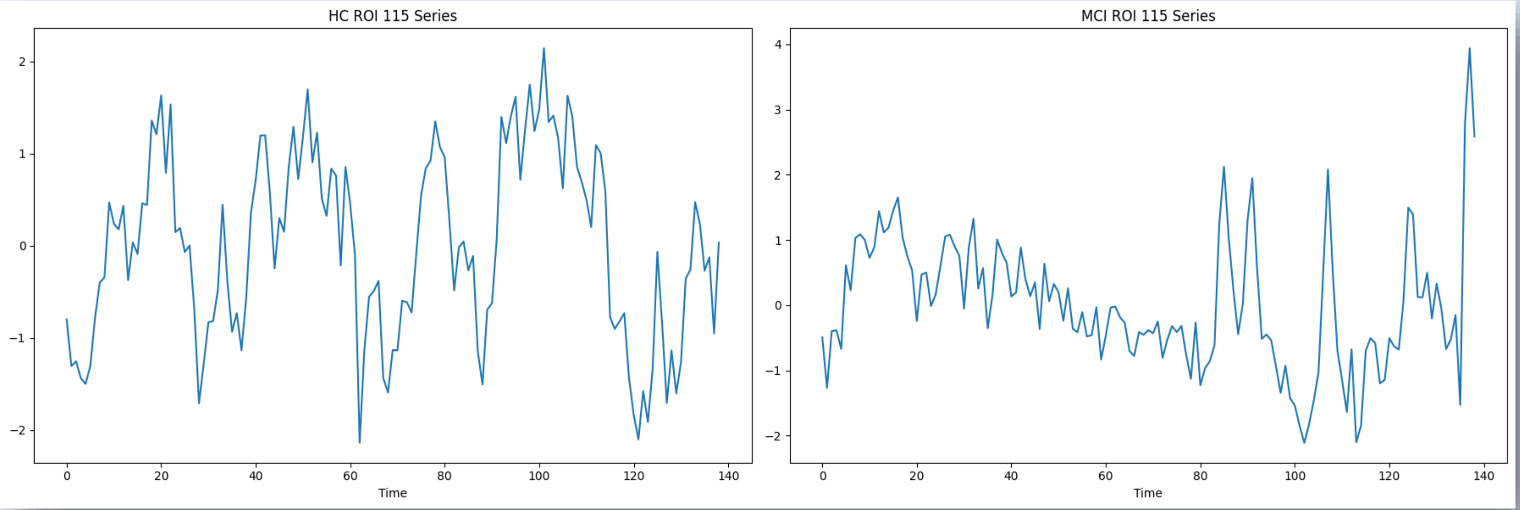
Output: Processed fMRI data with corresponding labels for subsequent analysis and modeling (2D CNN, ViT, LSTM).

Processed fMRI data

The resize **fMRI** objects (t,x,y,z) – (t,ROI)

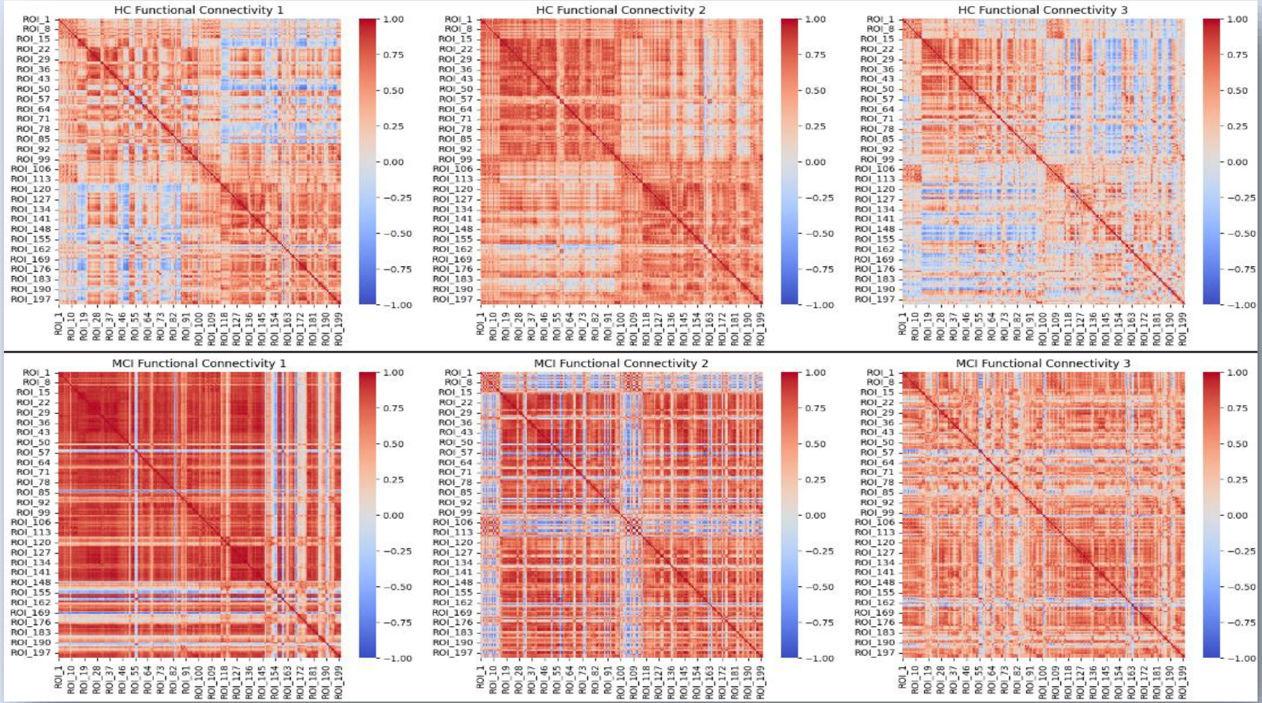


ROI series in HC vs MCI

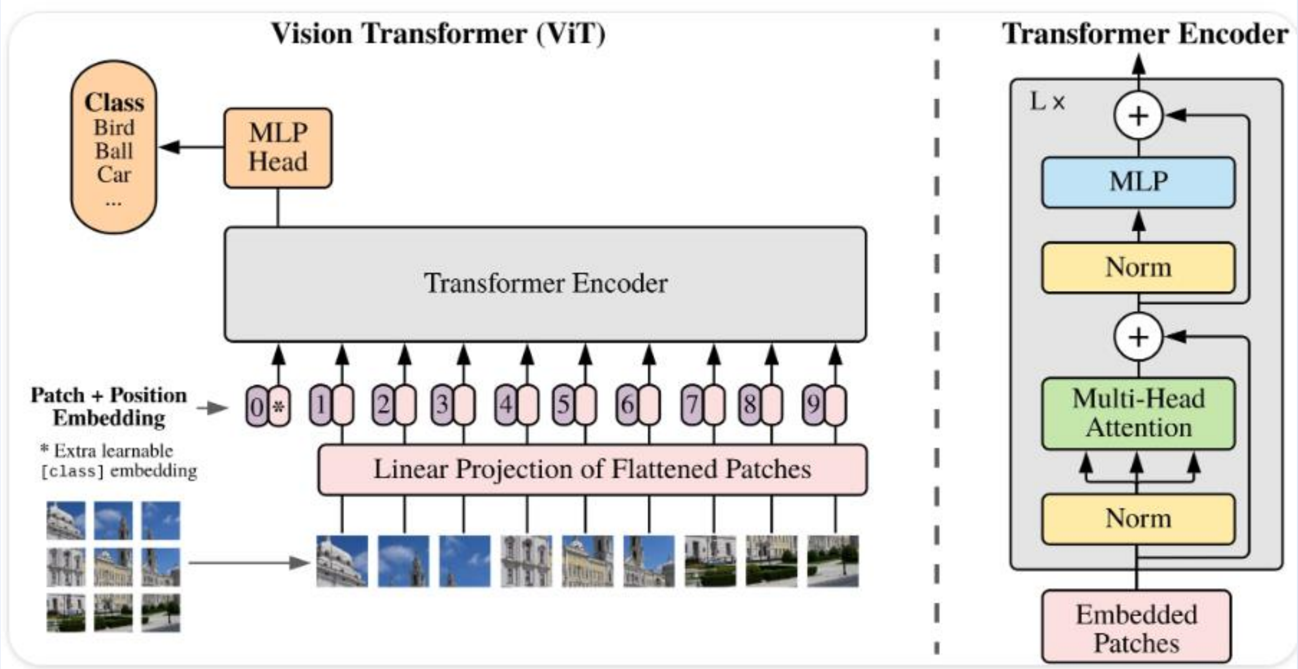


Functional Connectivity matrixes HC vs MCI

A functional connectivity matrix in fMRI images represents the functional connections between different brain regions (In our case – our ROIs).



Vision Transformer architecture



Vision Transformer model

Our vision transformer can be described as hybrid between a CNN and an embedding RNN self attention model.

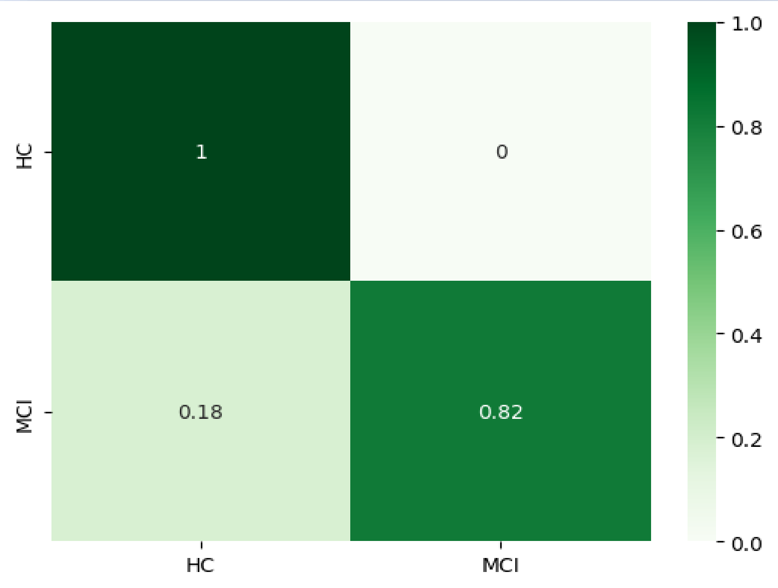
Hyperparameters:

Convolutional filters	128
Kernel size	3*3
Batch size	16
Embedding dimension	128
Attention heads	1
Transformer blocks	1
epochs	200 (stopped at 80)
Learning rate	0.001
loss	"CE"
optimizer	"Adam"
patience	30

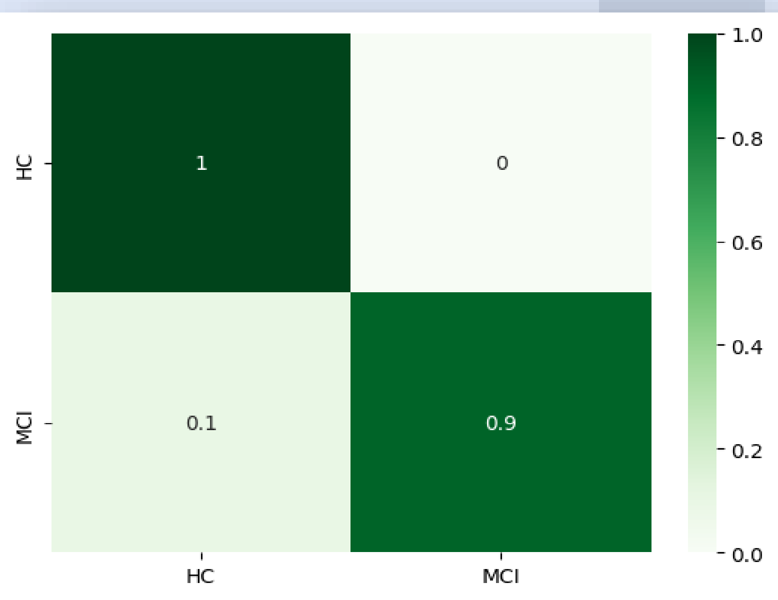
```
inputs = keras.Input(shape=input_shape)

x = layers.Conv2D(128, kernel_size=(3, 3), strides=(2, 2), padding='same', activation='relu')(inputs)
x = layers.BatchNormalization()(x)

embedding_dim = 128  # Embedding dimension
num_heads = 1  # Number of attention heads
transformer_units = [(embedding_dim * 2), (embedding_dim)]  # Feedforward network units
num_transformer_blocks = 1  # Adjust as needed

for _ in range(num_transformer_blocks):
    # Multi-head self-attention
    attention_output = layers.MultiHeadAttention(num_heads=num_heads, key_dim=embedding_dim)(x, x, x)
    attention_output = layers.LayerNormalization(epsilon=1e-6)(attention_output + x)

    # Feedforward network
    feedforward_output = layers.Conv2D(transformer_units[0], kernel_size=1, activation="relu")(attention_output)
    feedforward_output = layers.Conv2D(transformer_units[1], kernel_size=1)(feedforward_output)
    x = layers.LayerNormalization(epsilon=1e-6)(attention_output + feedforward_output)

x = layers.GlobalMaxPooling2D()(x)

outputs = layers.Dense(num_classes, activation="softmax")(x)

model = keras.models.Model(inputs=inputs, outputs=outputs)

model.compile(loss='sparse_categorical_crossentropy', optimizer='adam', metrics=['accuracy'])
```

Vistransformer

Test set

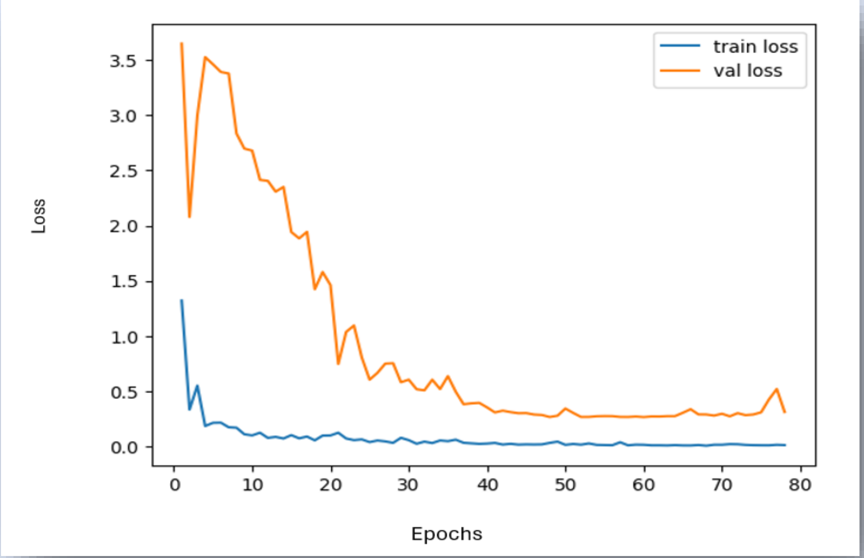


Validation set

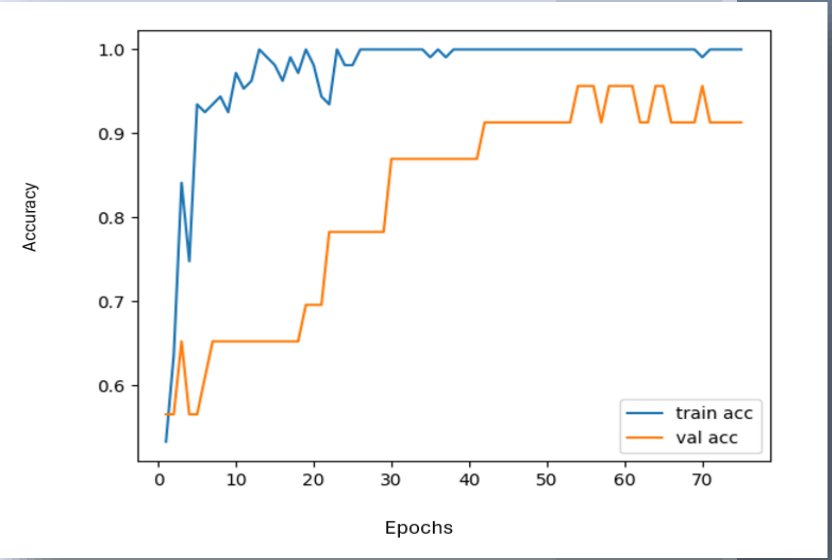


Vistransformer

Vistransformer Loss

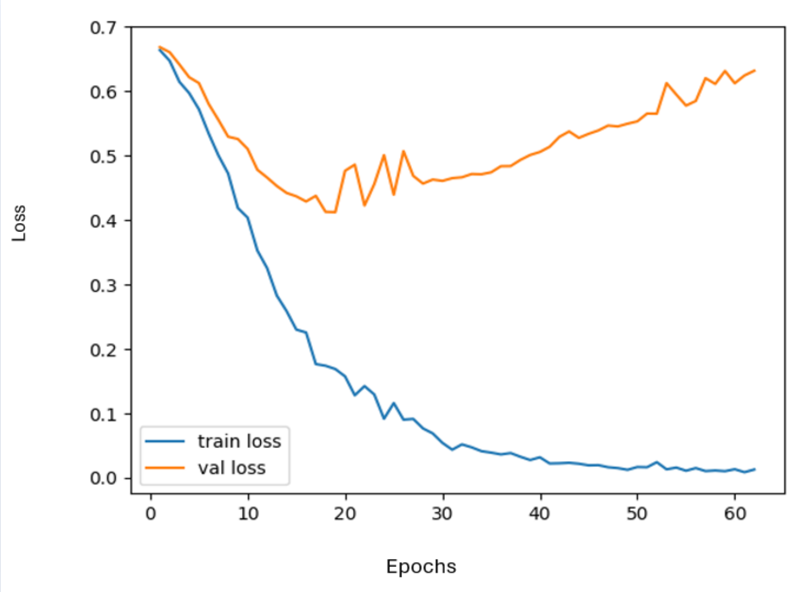


Vistransformer Accuracy

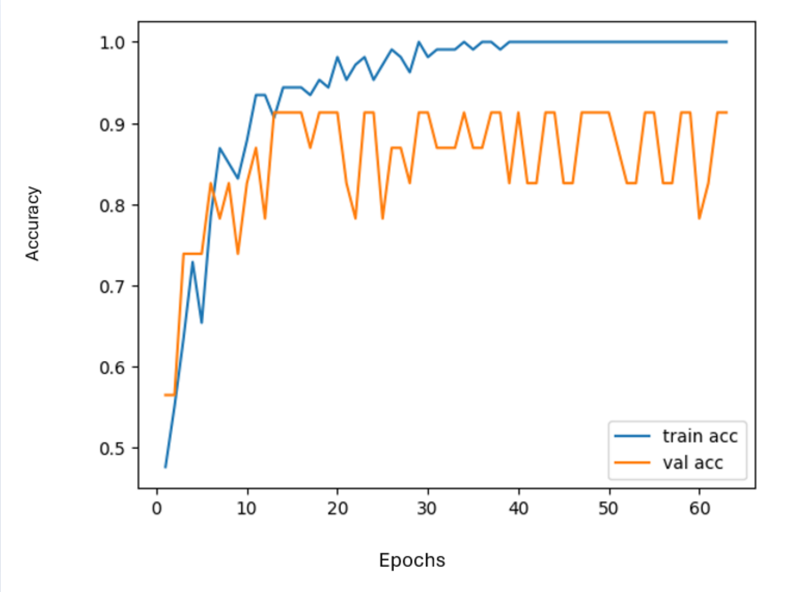


2D CNN Model

2D CNN Loss



2D CNN Accuracy

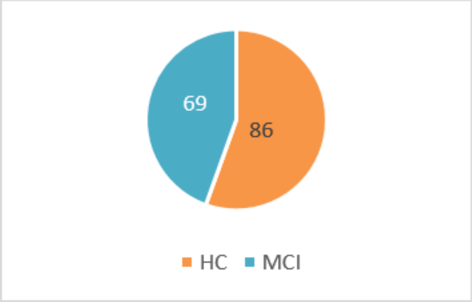


3D Anatomical CNN

We used 3D anatomical MRI data from Dr. Bolla (86 HC, 69 MCI, same participants).

Hyperparameters:

Convolutional filters	64,32
Kernel size	3*3*3
Batch size	16
dropout	0.3
Pool size	2*2*2
epochs	200 (stopped at 80)
Learning rate	0.001
loss	"CE"
optimizer	"Adam"
patience	30

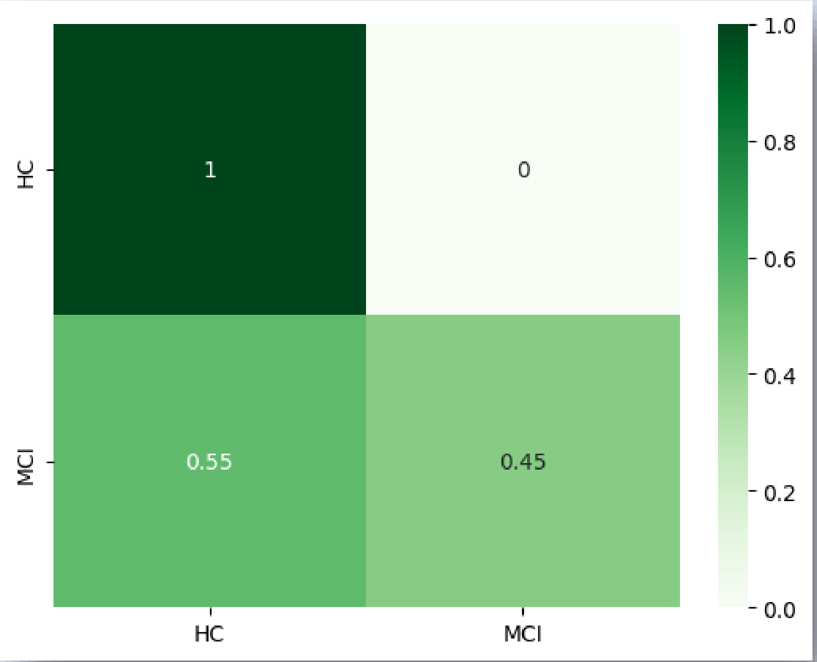


Validation accuracy: 86.9%
Test accuracy: 75%

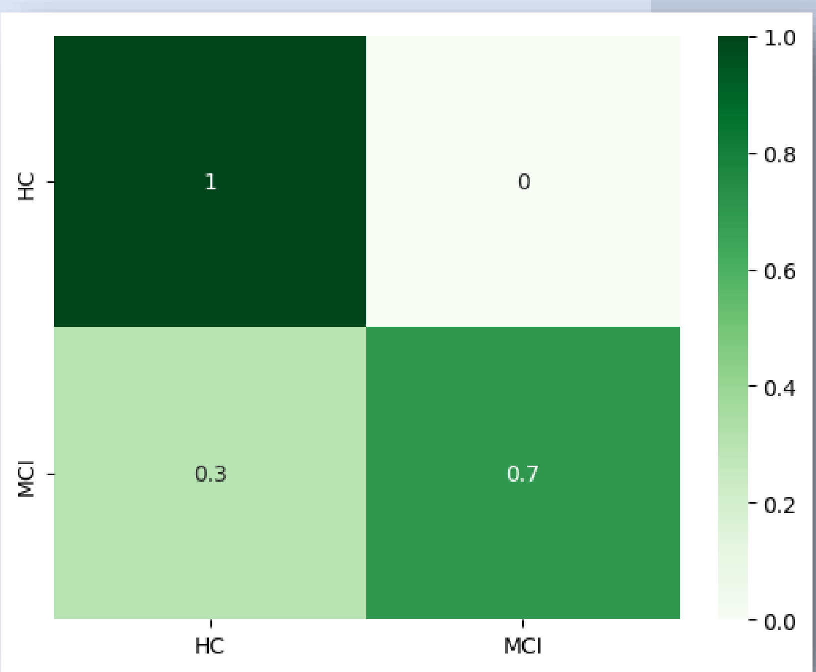
3D Anatomical CNN

Overfitting on HC class ,Perfect recall for HC, low for MCI.

Test set

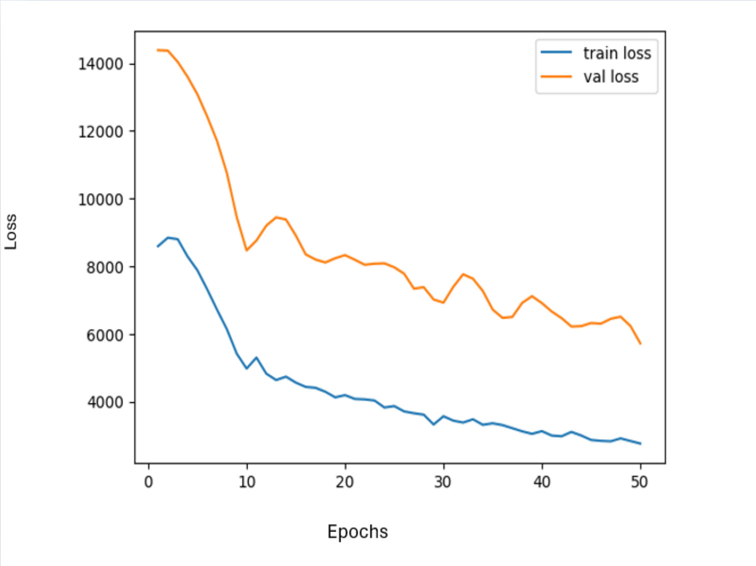


Validation set

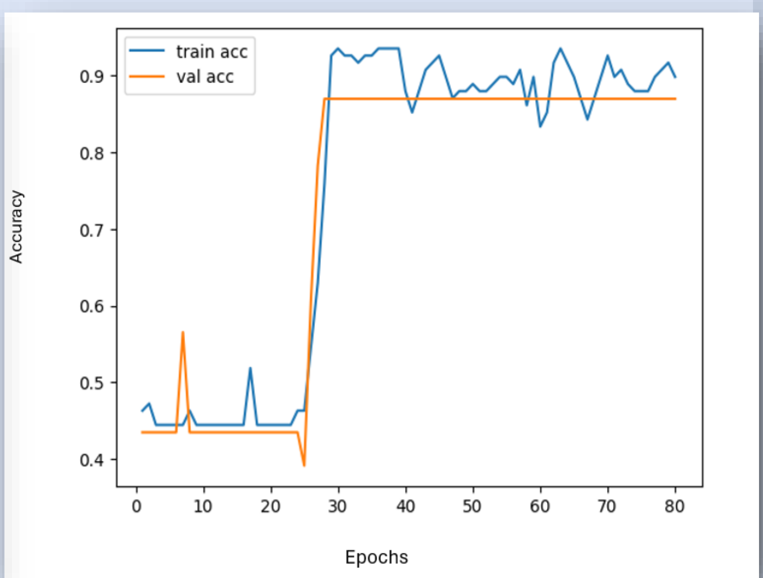


3D Anatomical CNN

CNN Loss



CNN Accuracy



Results

- Evaluated multiple models (ViT, 2D CNN, 3D CNN) on fMRI data.
- Vision Transformer (ViT) excelled due to its ability to capture spatial-temporal patterns.
- 2D CNN Performance Effective in capturing spatial features but prone to overfitting.
- fMRI data with time dimension and ViT's attention mechanism significantly improved performance compared to 3D anatomical CNN.

Model	Data	Validation Accuracy	Test Accuracy
Vision Transformer (ViT)	fMRI ROI x Time	95.6%	91.6%
2D CNN	fMRI ROI x Time	91.3%	95.8%
LSTM	fMRI ROI x Time	57%	57%
3D Anatomical CNN	3D Anatomical MRI	86.9%	75%

Conclusions

- Early and efficient prediction of AD and MCI using advanced neural network architectures.
- Functional connectivity analysis of fMRI data with special ROIs
- Utilized 3D CNNs and Vision Transformers for comprehensive brain analysis.
- Vision Transformer excelled in capturing spatiotemporal patterns, leading to improved classification accuracy.
- Continued research to enhance model architectures and data processing techniques.

Thank you!